

## CLASS XI · LECTURE 7

**Networks, Devices and the OSI Model**

Two periods · The highest-scoring diagram paper: OSI, LAN/WAN, topologies

**By the end of this lecture you can**

- Classify a network as LAN, MAN or WAN and state a limit of each.
- Compare bus, star and ring topologies.
- Place hub, switch, bridge, router and gateway in the right job.
- Draw the seven OSI layers in order and give one job of each.

**Why networks exist****DEFINITION Computer network**

A computer network is two or more computers, and other devices, linked so that they can share data, software and hardware such as a printer.

The gains are shared files, shared printers, a shared Internet connection, mail inside an office, and central backup. The costs are cabling or wireless equipment, a person to look after the network, and the new risk that a virus or an intruder can reach more than one machine.

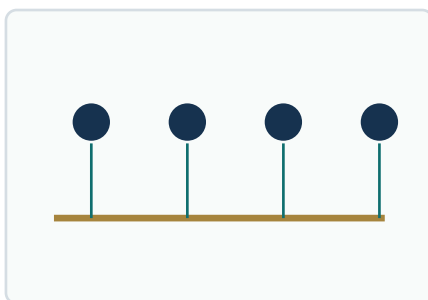
**Scale: LAN, MAN and WAN**

The Section C wording is “classify networks by their scale and limitation”. Scale means how far the network reaches. Limitation means what it cannot do, or what it costs you.

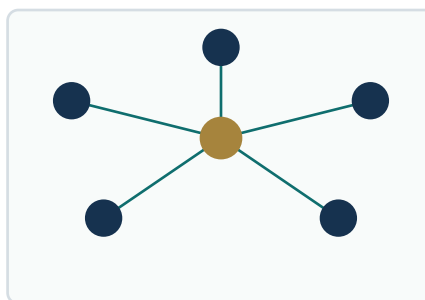
| Network                        | Scale                                                        | Owned by                                       | Limitation                                                                        |
|--------------------------------|--------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------|
| LAN, local area network        | A room, a lab, a building, sometimes a campus                | Usually one organisation                       | Does not cover a city or a country by itself                                      |
| MAN, metropolitan area network | A city, or several offices of one organisation across a city | Organisation or a city provider                | More costly than a LAN; not worldwide                                             |
| WAN, wide area network         | A country or the world. The Internet is the largest WAN      | Often uses lines rented from telecom companies | Slower and less private than a LAN unless extra protection is added; monthly cost |

**Topologies**

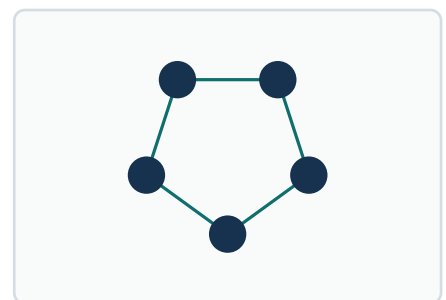
A topology is the shape of the connections, not the brand of the cable. Three shapes are enough for the paper if you can also name mesh, tree and hybrid in one line each.



Bus



Star



Ring

| Topology | Shape                                                               | Advantage                                                      | Weak point                                                                     |
|----------|---------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------|
| Bus      | One backbone cable. Devices tap onto it                             | Cheapest. Least cable. Easy to set up a small lab              | A break in the backbone stops the network. More collisions as you add machines |
| Star     | Each device has its own cable to a central hub or switch            | A broken spoke affects only one device. Easy to find the fault | If the centre fails, everything stops. More cable                              |
| Ring     | Each device joins the next in a loop. A token or frames pass around | Orderly access. No central hub in a pure ring                  | A break in the ring can stop traffic. Slower to extend                         |
| Mesh     | Many devices connected to many others                               | More than one path, so one break is survivable                 | Expensive. A full mesh needs a huge number of links                            |
| Tree     | Stars joined under a root, like a hierarchy                         | Matches a building with floors and departments                 | The root is a single weak point                                                |
| Hybrid   | A mixture, such as star of buses                                    | Fits a real campus                                             | Harder to design                                                               |

#### EXAM TIP Cheapest topology

The bus is the cheapest of the common topologies because one cable is shared and no central switch is required. The star is easier to maintain and is what modern switched LANs actually look like. If the question says cheapest, write bus. If it says most fault tolerant, write mesh.

## Devices, and the similar / different trick

BIEK pairs two ideas. A device that joins **two similar networks** is a **bridge**. A device that allows communication between **two different networks** is a **gateway**. Learn that pair for the wording of the paper, and then learn the fuller list so a long answer is not only two names.

| Device   | What it does                                                                        | Where it sits                                    |
|----------|-------------------------------------------------------------------------------------|--------------------------------------------------|
| NIC      | Connects one computer to the cable or the wireless network. Has a hardware address  | Inside or plugged into each computer             |
| Repeater | Rebuilds a weak signal so the cable can be longer                                   | Physical                                         |
| Hub      | A multiport repeater. Sends an incoming frame out of every other port               | Physical. A shared bus in a box                  |
| Switch   | Sends a frame only to the port that leads to the destination address                | Data link. The centre of a modern star           |
| Bridge   | Joins two similar network segments and filters traffic by hardware address          | Data link. The paper's "similar networks" device |
| Router   | Forwards packets from one network to another using IP addresses and a routing table | Network layer                                    |
| Gateway  | Connects networks that use different rules or protocols and translates between them | The paper's "different networks" device          |
| Modem    | Modulates and demodulates so digital data can use an analog line                    | At the end of a telephone or broadband link      |

In a fuller answer you may say that a router joins different IP networks, while a gateway is the device the paper names when the networks differ in protocol. Do not call a bridge a router. A bridge does not choose a path across the Internet. An NIC is not a topology.

## Standards and the OSI model

Standards exist so that equipment from different makers can work together. **ISO** is the International Organization for Standardization. **IEEE** sets many LAN standards, including the Ethernet family. **ITU** works on telecommunication standards. You need the names more than their history.

The **OSI model** has **seven** layers. It is a reference model: a way to teach and to design, not a single piece of software. Learn the order from the bottom, because cables are at the bottom and the user is at the top. A

sentence you can chant: **Please Do Not Throw Sausage Pizza Away** — Physical, Data link, Network, Transport, Session, Presentation, Application.

| Layer          | Job in one sentence                                                               | Example to quote            |
|----------------|-----------------------------------------------------------------------------------|-----------------------------|
| 7 Application  | The service the user actually touches                                             | HTTP, FTP, SMTP, a browser  |
| 6 Presentation | Format, encryption and compression so the two sides agree on the look of the data | ASCII, encryption           |
| 5 Session      | Opens, manages and closes a conversation                                          | Login session, checkpoints  |
| 4 Transport    | End-to-end delivery, and splitting a message into segments                        | TCP, UDP                    |
| 3 Network      | Logical addressing and choosing a path                                            | IP address, router          |
| 2 Data link    | Frames, hardware addresses, and the hop to the next device                        | MAC address, switch, bridge |
| 1 Physical     | The bits on the wire or the radio: voltages, light, connectors                    | Cable, hub, NIC pins        |

Figure 7.1 OSI layers from the user down to the cable. Draw them as a stack, layer 7 on top.

TCP/IP is the family the Internet actually runs. It is often drawn in four layers: application, transport, internet, and network access. You do not need to map every OSI layer onto TCP/IP unless a question asks. You do need to say that **TCP/IP** is the protocol suite of the Internet, and that a **protocol** is a set of rules.

#### EXAM TIP How to draw OSI for 10 marks

Draw seven stacked boxes, Application at the top, Physical at the bottom, numbered. Under the drawing, write one job for each layer. Add one example on Application (HTTP), Transport (TCP), Network (IP) and Data link (switch). Stop. A novel about the history of ISO does not add marks.

## Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

### Multiple choice

1. The OSI model contains:

- A) 5 layers
- B) 6 layers
- C) 7 layers
- D) 4 layers

2. The cheapest common topology is:

- A) Mesh
- B) Star
- C) Bus
- D) Fully connected

3. The device that connects two similar networks is a:

- A) Gateway
- B) Bridge
- C) Plotter
- D) Compiler

### Short answers

**S1.** Define the LAN device used to allow communication between two different networks.

**S2.** Differentiate broadcast and point-to-point communication.

### Detailed answer

**L1.** Draw and explain the OSI reference model.

**L2.** What is a computer network? Classify networks by scale and state a limitation of each.

### Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

**MCQ 1.** C. Physical through Application. TCP/IP is the one often drawn with four.

**MCQ 2.** C. One shared cable and no central switch.

**MCQ 3.** B. Gateway is the paper's answer for different networks.

**S1.** A gateway connects networks that are different, often because they use different protocols, and translates between them so data can pass. In the same family of questions, a bridge joins two similar networks, and a router forwards packets between networks by using addresses and a routing table.

**S2.** Point-to-point communication links one sender with one receiver, as in a dedicated cable between two computers or a telephone call. Broadcast communication sends one transmission to many receivers at the same time, as in radio or a hub repeating a frame to every port.

**L1.** OSI is a seven-layer reference model for network communication, with the user at the top and the cable at the bottom. Application provides services such as HTTP, FTP and SMTP. Presentation handles format, compression and encryption. Session opens and closes the dialogue. Transport gives end-to-end delivery and splits data into segments; TCP works here. Network provides logical addressing and routing; IP and routers work here. Data link makes frames, uses hardware addresses, and includes switches and bridges. Physical transmits bits as voltages or light over cable or radio. Draw the seven names in a stack, Application on top. Data passes down the sender's stack and up the receiver's stack.

**L2.** A computer network is a set of computers and other devices linked to share data and resources. By scale: a LAN covers a room, building or campus, is usually private, and cannot by itself span a city. A MAN covers a city, for example several branches of one college, and costs more than a LAN while still not covering a country. A WAN covers a large geographical area. The Internet is the largest WAN. A WAN often rents public lines, so it costs a monthly fee and is generally slower and more exposed than a LAN unless it is protected. Advantages common to all three are sharing of files, printers and data.